AI651 – Deep Learning for Space, Time and Graphs
Assignment 1 Report

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GitHub repository (public): `https://github.com/OsamaAtiq12/DL4STG-PA1`

1 Task 1 – Forecasting Across Heterogeneous Sensors

From a context of L=96 samples we forecast H=48 vibration readings (20 Hz). Stations 1–3 are established; Station 4 is held out. Products A/B/C run at different paces. All Task 1 numbers and figures below come from one fresh `PA1_PRESET=full` notebook run (data seed 0, model seed 0).

1.1 Response 1A – Ordering of forecasters

Before Output 1.1 I expected

$$\text{Period-routed ridge} \succ \text{Raw Attention} \succ \text{Shared ridge} \succ \text{Sensor-specific ridge}$$

on established stations, and the same on Station 4 except that sensor-specific ridge cannot be scored there (no Station 4 weights).

After Output 1.1 (validation RMSE, m/s^2):

Model	Established	Held-out
Period-routed ridge	**0.166**	**0.173**
Raw Attention	0.431	0.612
Shared ridge	0.767	0.876
Sensor-specific ridge	0.768	—

The measured order matches the prediction. Period routing wins on both populations because it conditions on operating pace. Shared and sensor-specific ridge are nearly tied on established stations; Raw Attention beats both but stays behind period routing.

1.2 Response 1B – Why the rule must depend on context

Output 1.2 (Station 1): Product A has period ≈16.1 and Product C has period ≈30.1. Shared ridge RMSE is 0.779 (A) and 0.448 (C); period-routed ridge drops to 0.232 and 0.150. One fixed shared matrix cannot handle both paces, so it compromises.

Output 1.3: the same Attention projections give different mixing weights across those two contexts (mean $|\Delta w|$=0.011, max 0.425). That is context-dependent mixing, not recovery of the true period. The forecasting rule must depend on the current window.

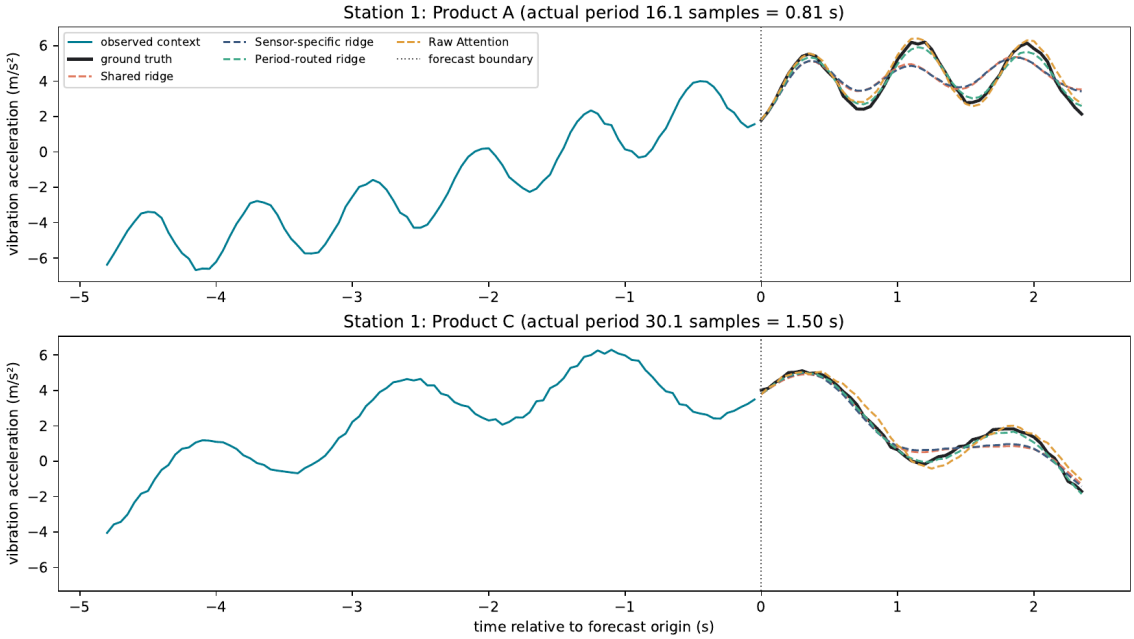


Figure 1: Output 1.2: same station, two products.

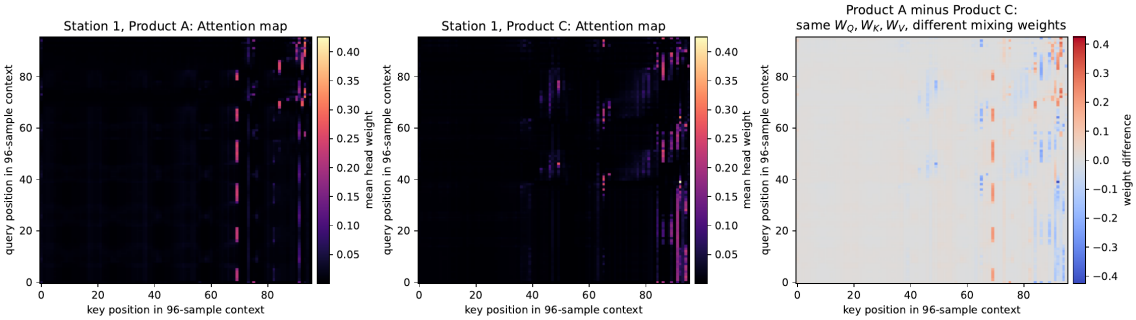


Figure 2: Output 1.3: Attention weight differences across contexts.

1.3 Response 2 – Decomposition

Selected width: **41** (Outputs 2.1–2.2). It gives the best oracle period recovery within one sample (0.998 vs. 0.687 on the raw signal). A centered moving average is a sensible bias here because the slow drift changes more slowly than the 14–36 sample vibration cycle, so a fixed smoother can peel trend from residual before mixing. Output 2.3 shows decomposition also lowers Attention RMSE (e.g. established Product A: 0.388 → 0.264).

Failure case: a sudden level jump or strong curvature inside one window can be mis-split by a fixed-width average. An alternative is a learned / causal trend head with adaptive bandwidth.

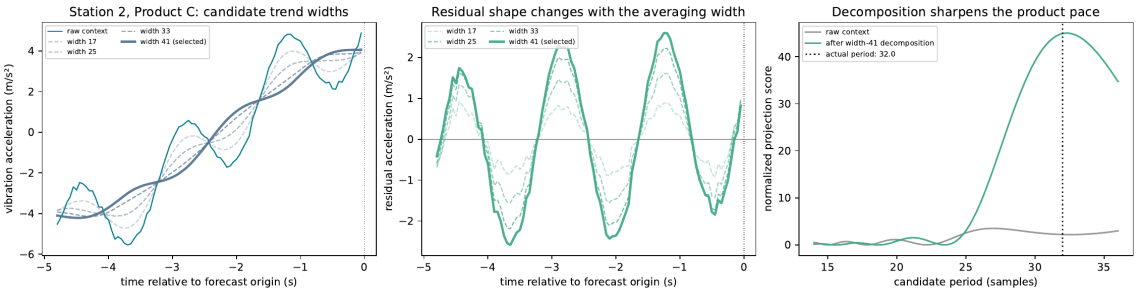


Figure 3: Output 2.1: candidate widths.

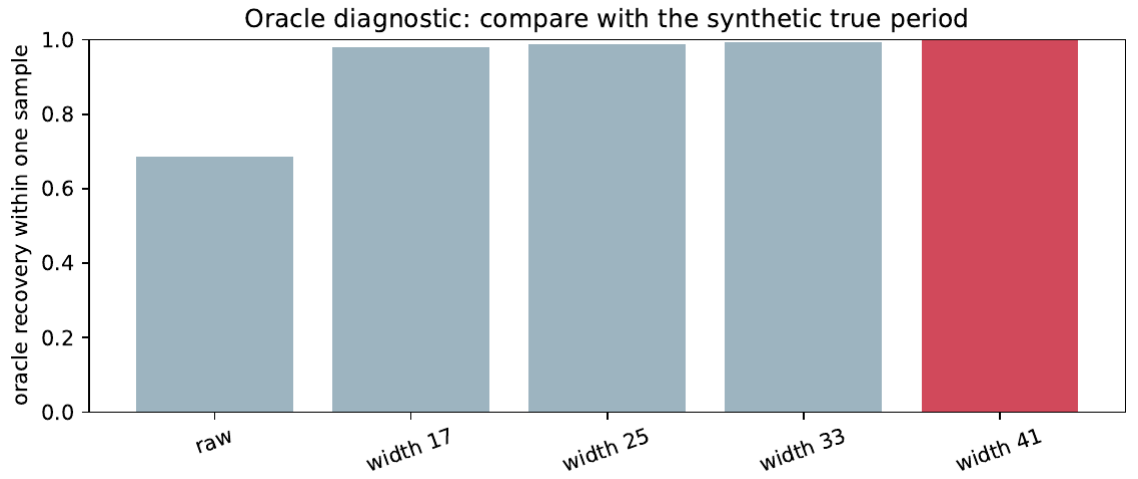


Figure 4: Output 2.2: oracle period recovery.

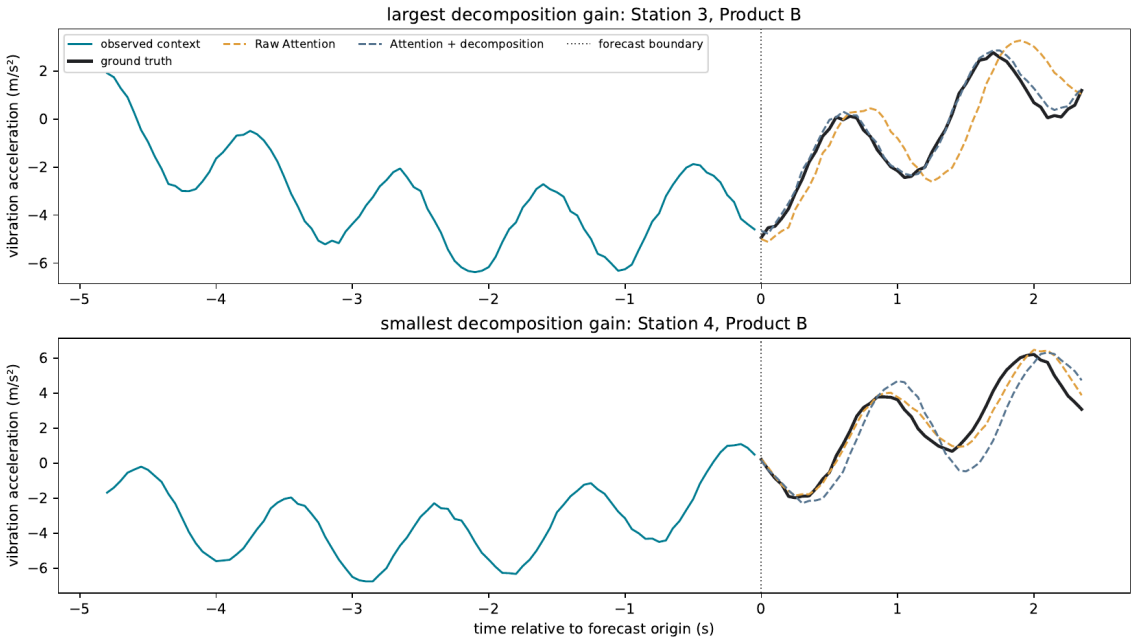


Figure 5: Output 2.3: Attention with vs. without decomposition.

1.4 Response 3 – Delay mixing

Output 3.2 (Product A, P≈16.13): strongest residual correlations are at delays 16 and 32 (≈0.896), i.e. P and $2P$. After removing the slow component these peaks are clearer, so delay mixing has cleaner evidence. Output 3.1 checks that FFT delay scores match direct circular correlation.

Delay mixing is the wrong bias for an aperiodic transient or a period that drifts inside one L=96 window, because top-K lag aggregation would mix mismatched histories.

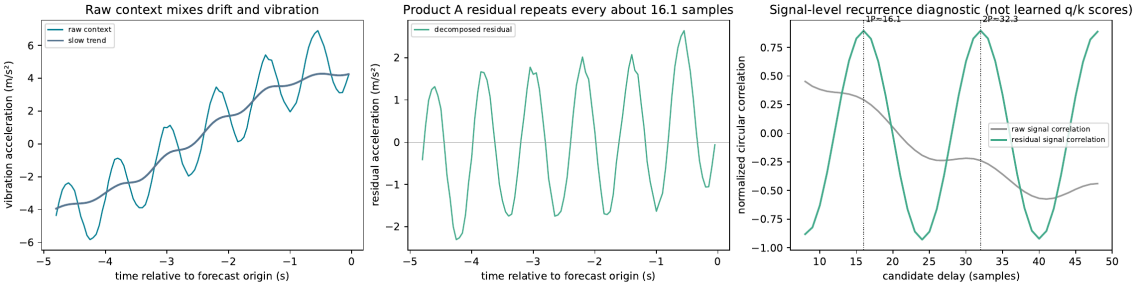


Figure 6: Output 3.2: recurrence diagnostic.

1.5 Response 4 – Compare and deploy

(a) Before Output 4.1. When the slow component is strong, I expected decomposition to help most: the trend head takes the ramp and the mixer sees a cleaner residual.

(b) Output 4.1–4.2. Among neural configs, Autoformer-inspired is best on established stations (RMSE 0.197) and on Station 4 (0.245), then Raw delay mixer (0.240 / 0.363), Attention+decomposition (0.296 / 0.430), and Raw Attention (0.431 / 0.612). Adding delay structure helps more than Attention alone; combining both switches is strongest among the neural models. Horizon error still grows with lead time (Output 4.2). Period-routed ridge remains best overall and cheapest (Output 1.1: 0.166 / 0.173).

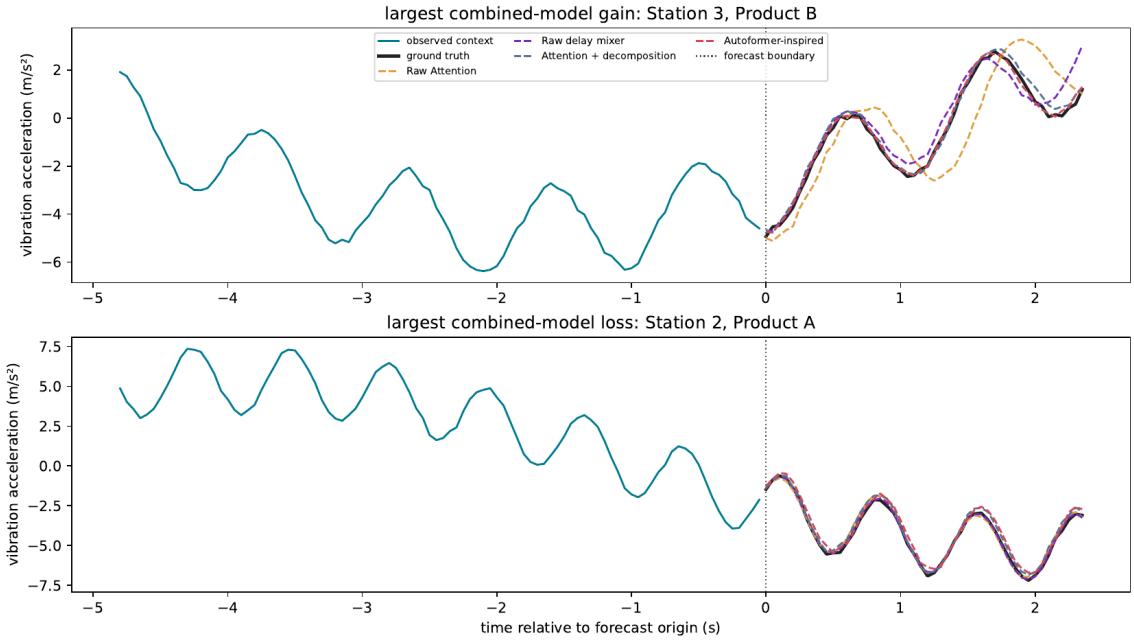


Figure 7: Output 4.1: neural comparison.

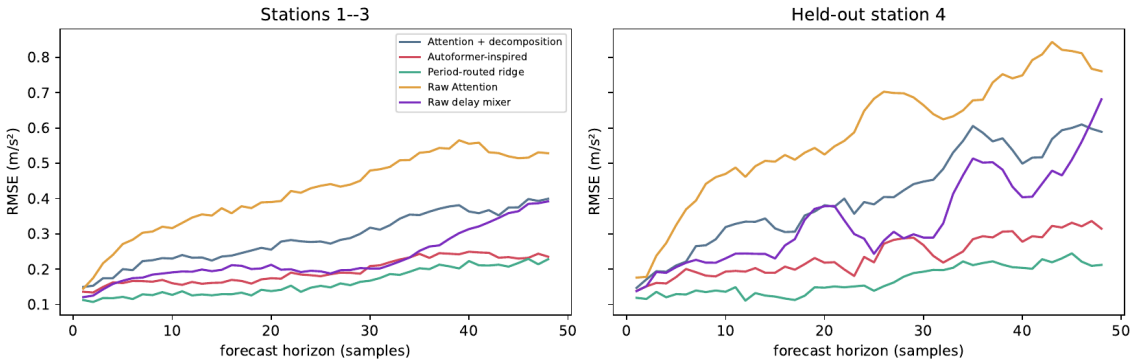


Figure 8: Output 4.2: horizon-wise error.

(c) Deployment.

Population	Model	Reason
Established	Period-routed ridge	Best val RMSE/MSE, low cost
Held-out Station 4	Period-routed ridge	Best held-out RMSE

Output 4.3 test RMSE: established 0.161; held-out 0.183.

2 Task 2 – Leaderboard Challenge

Compact Autoformer (decomposition + Auto-Correlation / delay aggregation): context 336, label 84, horizon 168, d=32, one encoder/decoder layer; AdamW; two seeds; final forecast averages the seeds after a short full-history refit.

Best validation RMSE:

Setting	Seed 0	Seed 1
Without external features	94.83	95.70
With external features	**62.06**	**64.07**

External features help, so the submission uses them. Declared P=46532, E=6.

AI Policy Compliance

Tools used: GitHub Copilot was used for code completion, syntax assistance, and concept understanding. All final code, numbers, and written Responses were checked and edited by me.